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SYSTEM STATUS

0.772

TEMPORAL AUC (REALISTIC)

Cross-val AUC: 0.973 (optimistic)

✓ Model loaded

SmartDaaS

AI-Powered HIV Programme Intelligence · Operational Analytics · Decision Support

v1.0 · Decision-Support Platform · Pilot-Ready

v1.2.0 – 15 May 2026 Local recalibration engine · DUA acknowledgement gate · Extended audit trail · Supabase event logging

SMART DISEASE - AS - A - SERVICE

AI-powered intelligence for HIV programmes

Identify high-risk patients. Detect underperforming facilities. Quantify avoidable costs. Drive better outcomes.

SmartDaaS combines patient-level risk prediction with facility intelligence and economic insights into one platform. Built on programme data from **27,288 HIV patients on ART** (Nigerian national HIV programme — discovery cohort) with local recalibration support for any country context. Designed for PEPFAR implementing partners, Global Fund grantees, and national HIV programme offices across sub-Saharan Africa.

0.963

CROSS-VAL AUC

0.772

TEMPORAL AUC

87.3%

SENSITIVITY

27K

TRAINING RECORDS
(DISCOVERY COHORT)

KEY CAPABILITIES



Patient Risk Engine

Upload your data and instantly get individual risk scores with clear SHAP explanations. Know who needs attention before they are lost to follow-up.



Facility Intelligence

Identify care gaps and structural drivers of poor performance. Risk-adjusted benchmarking separates facility quality from patient case-mix.



Economic Impact

Calculator

Quantify excess poor outcomes and potential cost savings. Three scenario presets built on published PEPFAR cost data.



Intervention Engine

Specific, tiered recommendations aligned with PEPFAR MER indicators TX_CURR, TX_PVLS, TX_ML. CHW outreach, clinical review, retention tracing.

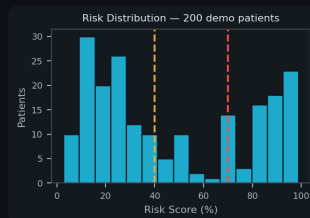


Executive Reports

Professional one-click PDF reports for donors and leadership. Ready to present at quarterly review meetings.

LIVE PLATFORM PREVIEW

See SmartDaaS working — live demo data, no upload required.



70

HIGH RISK $\geq 70\%$

23

MEDIUM RISK

107

LOW RISK

Top 5 highest risk patients:

	Risk %	Tier
PT-0000	98.9	● HIGH
PT-0001	98.6	● HIGH
PT-0002	98.5	● HIGH
PT-0003	98.3	● HIGH
PT-0004	98.3	● HIGH

Ready to see it in action?

Upload your patient data or try with demo data → go to  **Patient Risk** in the sidebar

Or see the  **Pilot Model** page to learn about our Shadow Analytics Implementation Programme

RESEARCH FOUNDATION

Paper 1 — Under review at npj Digital Medicine Chinthala LK. *Real-World Validation of Machine Learning Models for HIV Treatment Adherence Prediction and Care Gap Quantification: Validation on 27,288 Nigerian HIV Programme Records with Temporal Holdout, SHAP Explainability, and Decision Curve Analysis*. 2026. 
Code: github.com/Kchinthala15/smartdaas-hiv-validation

Paper 2 — Under preparation for submission Chinthala LK. *Facility-Level Structural Drivers of HIV Treatment Outcomes: A Multi-Level Analysis of 27,288 Patients from a Nigerian HIV Programme and Implications for PEPFAR and Global Fund Programming*. 2026.

DOIs will be added upon publication.

Target users: HIV programme managers · PEPFAR implementing partners · Global Fund grantees · National HIV programme officers · MoH analytics teams

ABOUT THE DEVELOPER

Lakshmi Kalyani Chinthala

Founder, SmartDaaS · Ageno School of Business, Golden Gate University, San Francisco, CA. Specialising in machine learning applications for global health programme management, with a focus on HIV/AIDS in sub-Saharan Africa.

DATA PRIVACY & HANDLING POLICY

Privacy by Design

No data storage: Patient data is processed within your browser session and never stored, transmitted externally, or retained after session ends.

No third-party sharing: Your programme data is never shared with any third party.

Pilot deployments: A formal Data Use Agreement (DUA) is established with all pilot partners before any data is ingested. All pilots operate under Nigeria's NDPR and relevant PEPFAR data governance frameworks.

 SmartDaaS is a decision-support platform designed to assist HIV programme teams. All outputs should be reviewed by qualified programme and clinical staff. Local validation on your programme's data is strongly recommended before full operational use.

0.963

CROSS-VAL AUC

0.772

TEMPORAL AUC

87.3%

SENSITIVITY

27K

TRAINING RECORDS
(DISCOVERY COHORT)

WHAT SMARTDAAS DOES



Patient Risk

Upload patient records.
Get individual risk scores with SHAP explanations.
Export prioritised patient lists.



Facility Intel

Identify underperforming facilities. Care-gap heatmaps. Structural drivers after case-mix adjustment.



Economics

Quantify excess poor outcomes. Calculate avoidable programme costs. Build the case for investment.

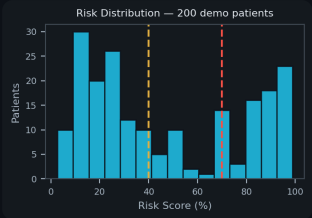


Actions

Tier-specific intervention recommendations aligned with PEPFAR MER indicators TX_CURR, TX_PVLS, TX_ML.

LIVE PLATFORM PREVIEW

See SmartDaaS working — live demo data, no upload required.



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MEDIUM RISK

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PT-0002	98.5	HIGH
PT-0003	98.3	HIGH
PT-0004	98.3	HIGH

QUICK START

1. Go to 📄 **Sample Data** — download the CSV template
2. Fill it with your patient data (or use demo data)
3. Go to 🏠 **Patient Risk** — upload and get risk scores instantly
4. Export your high-risk patient list for follow-up

RESEARCH FOUNDATION

Paper 1 — Under review at npj Digital Medicine Chinthala LK. *Real-World Validation of Machine Learning Models for HIV Treatment Adherence Prediction and Care Gap Quantification: Validation on 27,288 Nigerian HIV Programme Records (Discovery Cohort)*. 2026. 🔗 Code: github.com/Kchinthala15/smartdaas-hiv-validation

Paper 2 — Under preparation for submission Chinthala LK. *Facility-Level Structural Drivers of HIV Treatment Outcomes: A Multi-Level Analysis of 27,288 Patients from a Nigerian HIV Programme and Implications for PEPFAR and Global Fund Programming*. 2026.

DOIs will be added upon publication. Analysis code openly available on GitHub.

Target users: HIV programme managers · PEPFAR implementing partners · Global Fund grantees · National HIV programme officers · MoH analytics teams

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Lakshmi Kalyani Chinthala

Founder, SmartDaaS · Ageno School of Business, Golden Gate University, San Francisco, CA. Specialising in machine learning applications for global health programme management, with a focus on HIV/AIDS in sub-Saharan Africa.

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DATA PRIVACY & HANDLING POLICY

🔒 Privacy by Design

No data storage: Patient data uploaded to SmartDaaS is processed entirely within your browser session and is never stored, transmitted to external servers, or retained after your session ends.

No third-party sharing: Your programme data is not shared with any third party, including the developer.

Session-only processing: All risk scores, SHAP explanations, and reports are generated in real time and exist only during your active session.

For pilot deployments: A formal Data Use Agreement (DUA) will be established with all pilot partners before any programme data is ingested. All pilots operate under applicable data protection regulations including Nigeria's NDPR and relevant PEPFAR data governance frameworks.

Platform notice: SmartDaaS v1.0 is a decision-support platform designed for HIV programme intelligence and operational analytics. It is not intended to replace clinical judgment. All outputs should be reviewed and validated by qualified programme teams prior to operational use. patient treatment decisions without clinical review and appropriate regulatory approval.

Interested in a Shadow Analytics Pilot?

SmartDaaS is seeking one implementing partner for a 6-month analytical pilot. You provide historical programme data. We deliver patient risk intelligence and facility performance reports.

✉ chinthalakalyani1@gmail.com

github.com/Kchinthala15/smartdaas-hiv-validation

👉 See the 📄 **Pilot Model** page for full details

⚠ **Decision-support platform — not a clinical tool.**
SmartDaaS v1.0 is designed for HIV programme intelligence and operational analytics. It is not intended to replace clinical judgment or function as an autonomous clinical decision-making system. All outputs should be reviewed by qualified programme teams. Local validation is required before deployment in real-world programme environments.

